

The Urban PlugIn

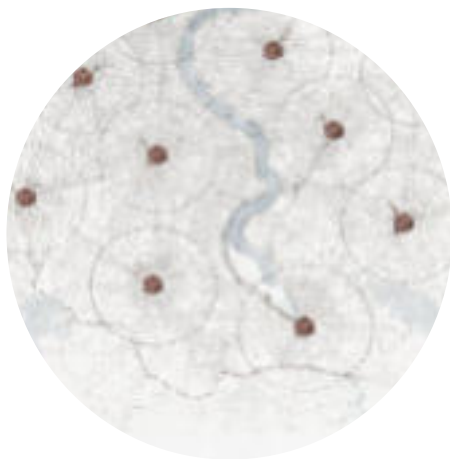
CONTENT



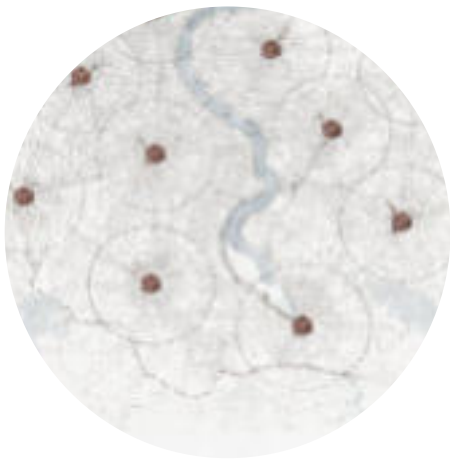
PRAGUE LOGISTICS



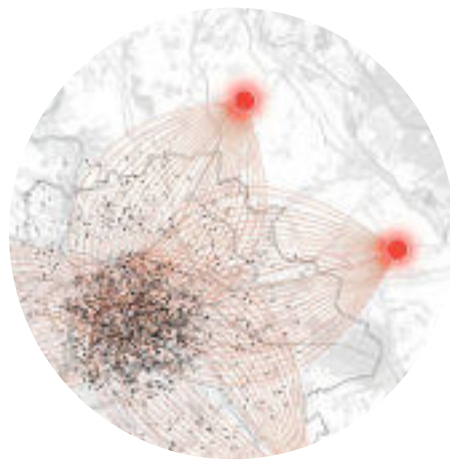
THE CURRENT SYSTEM



THE EXAMPLE SYSTEM



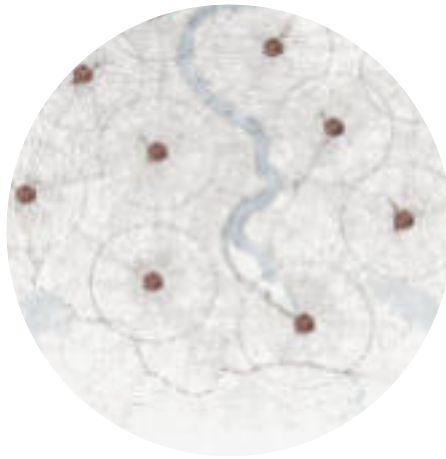
THE CHANGED SYSTEM



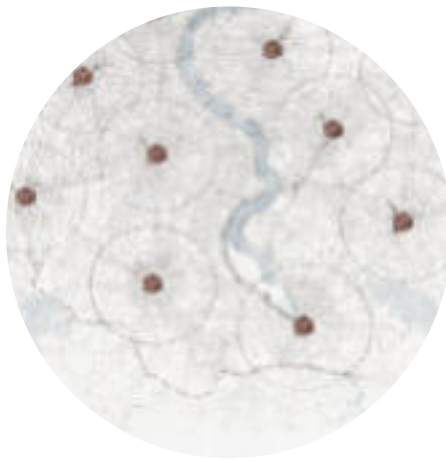
PRAGUE URBAN STUDY



DESIGN CONCEPT



DESIGN INTEGRATION



KPI

PRAGUE LOGISTICS

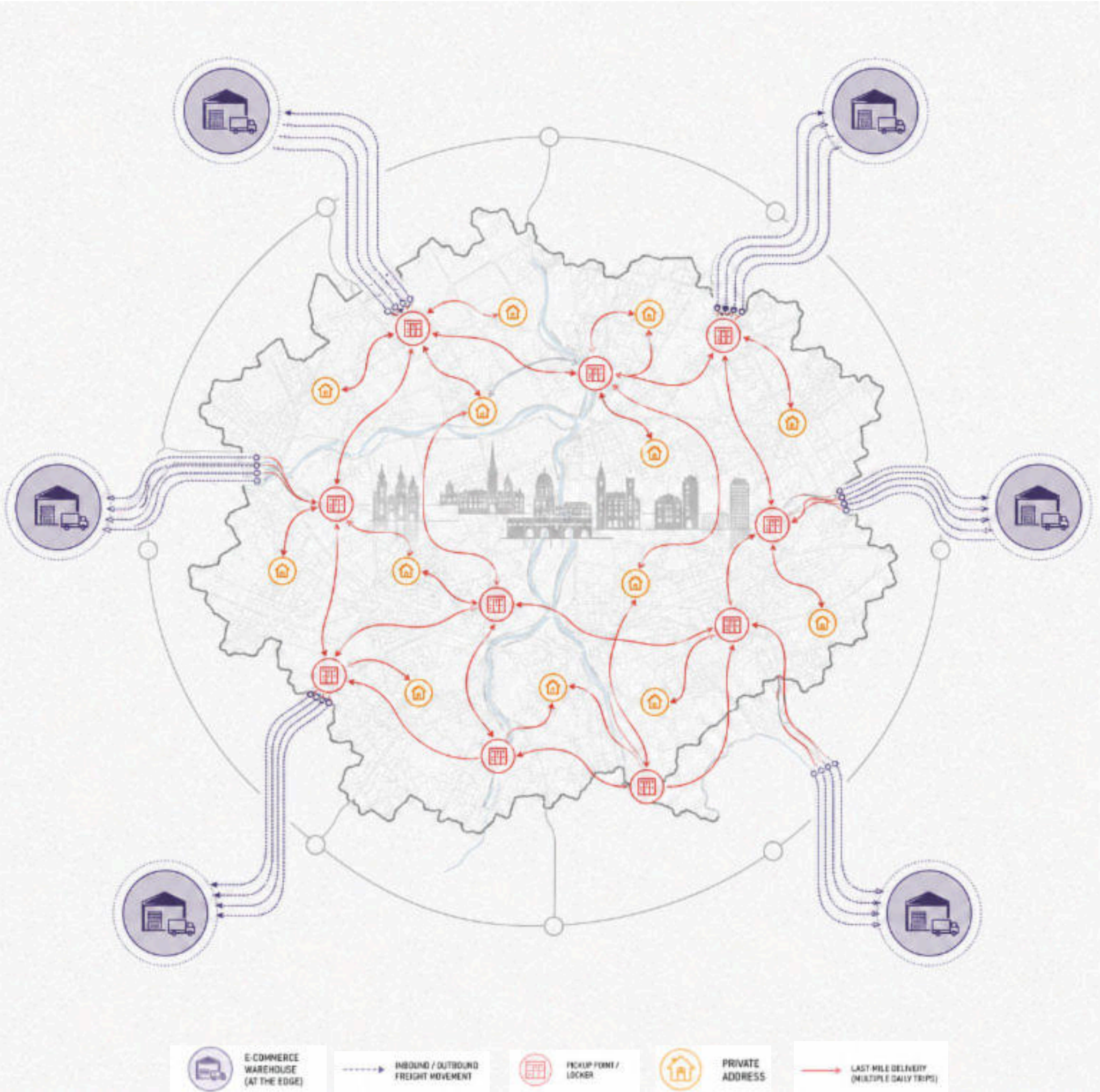
What is the current situation of Prague ?

THE CURRENT SYSTEM

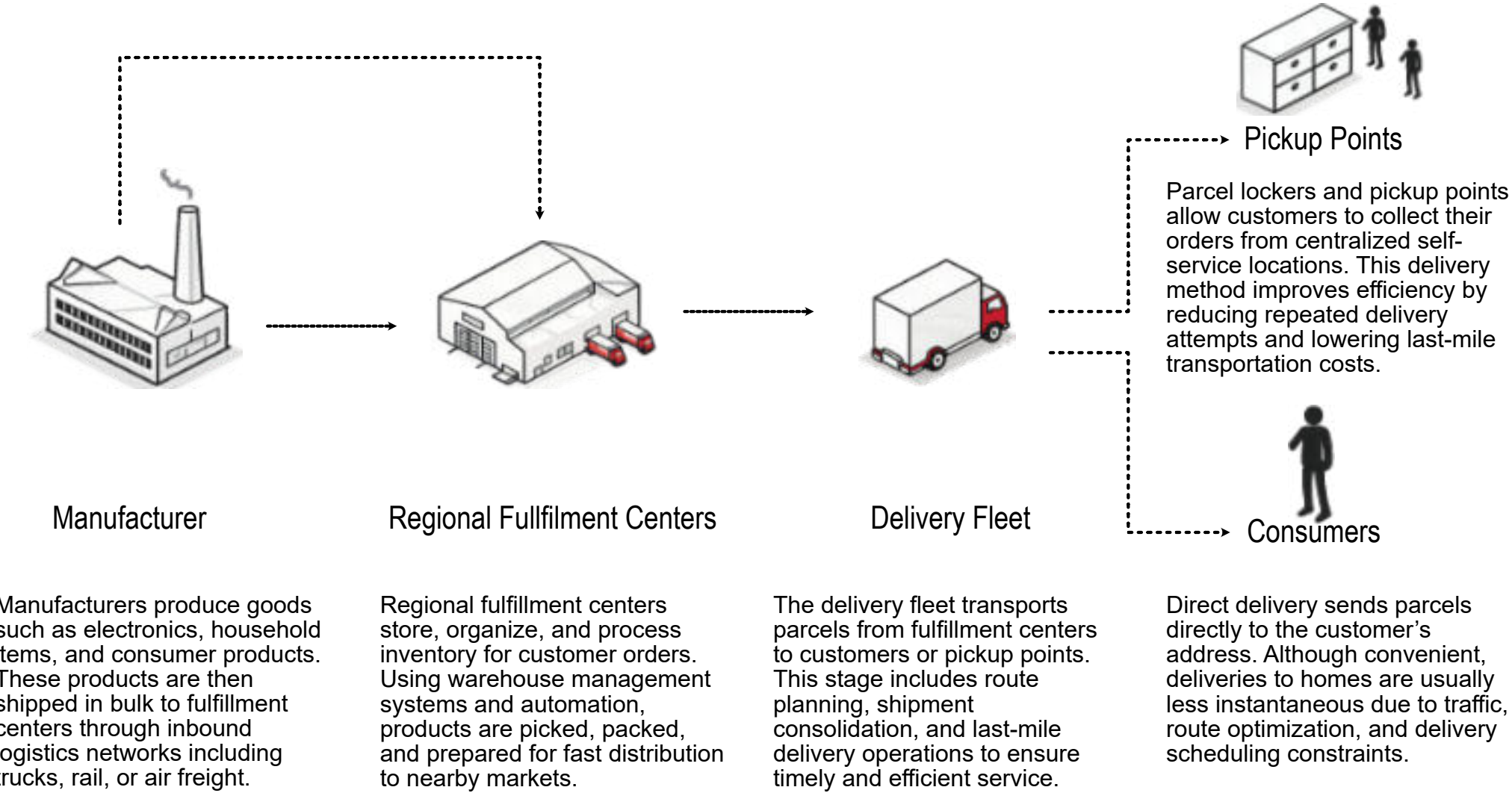
DECENTRALISED AND INSUFFICIENT

Prague’s current e-commerce logistics system operates through a decentralised network of large distribution and fulfilment centres located along the city’s outer edge, strategically positioned near major highway corridors such as the D1, D5, D8, and Prague Ring Road. Major logistics zones have developed around areas including Horní Počernice, Hostivař–Stěboholy, Čakovice, and Chýně/Chrásťany, where companies such as Rohlik.cz, Alza, DHL, PPL, DPD, GLS, Amazon distribution partners, and Zásilkovna operate warehouses, sorting facilities, and delivery hubs. From these peripheral centres, goods are transported into the city through fragmented last-mile delivery networks, often resulting in multiple companies servicing the same neighbourhoods simultaneously.

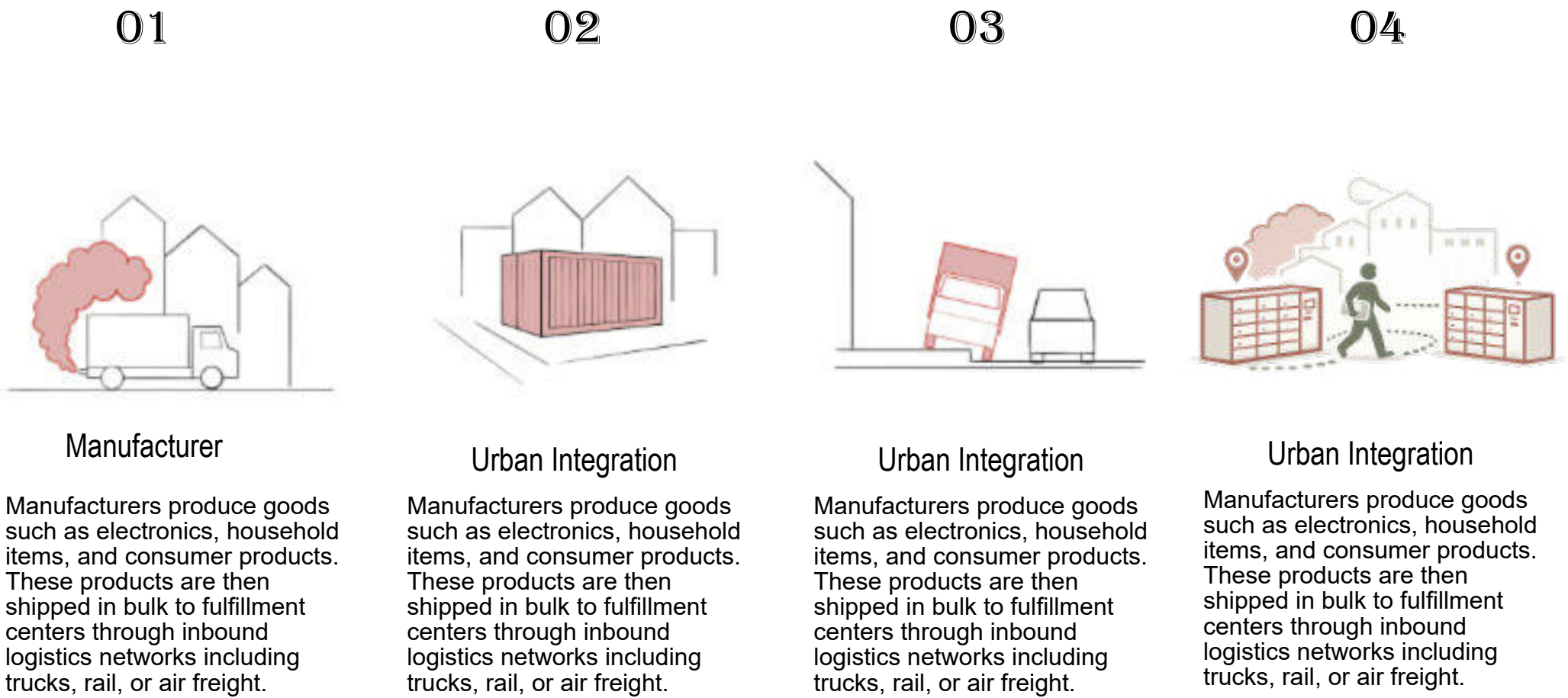
While this system supports the rapid growth of e-commerce and next-day delivery expectations, it also generates increasing pressure on Prague’s urban infrastructure through traffic congestion, duplicated delivery routes, emissions, and the intensive occupation of public streets by freight vehicles. The concentration of logistics infrastructure outside the city reinforces a strong dependency on road-based transportation and long delivery flows into the urban core. As online retail continues to expand, Prague reveals the limitations of the current logistics model and becomes a critical site for exploring more integrated, efficient, and sustainable urban delivery systems.



1.1 HOW THE CURRENT SUPPLY CHAIN WORKS



1.2 KEY PRBLEMS WITH THE CURRENT SUPPLY CHAIN



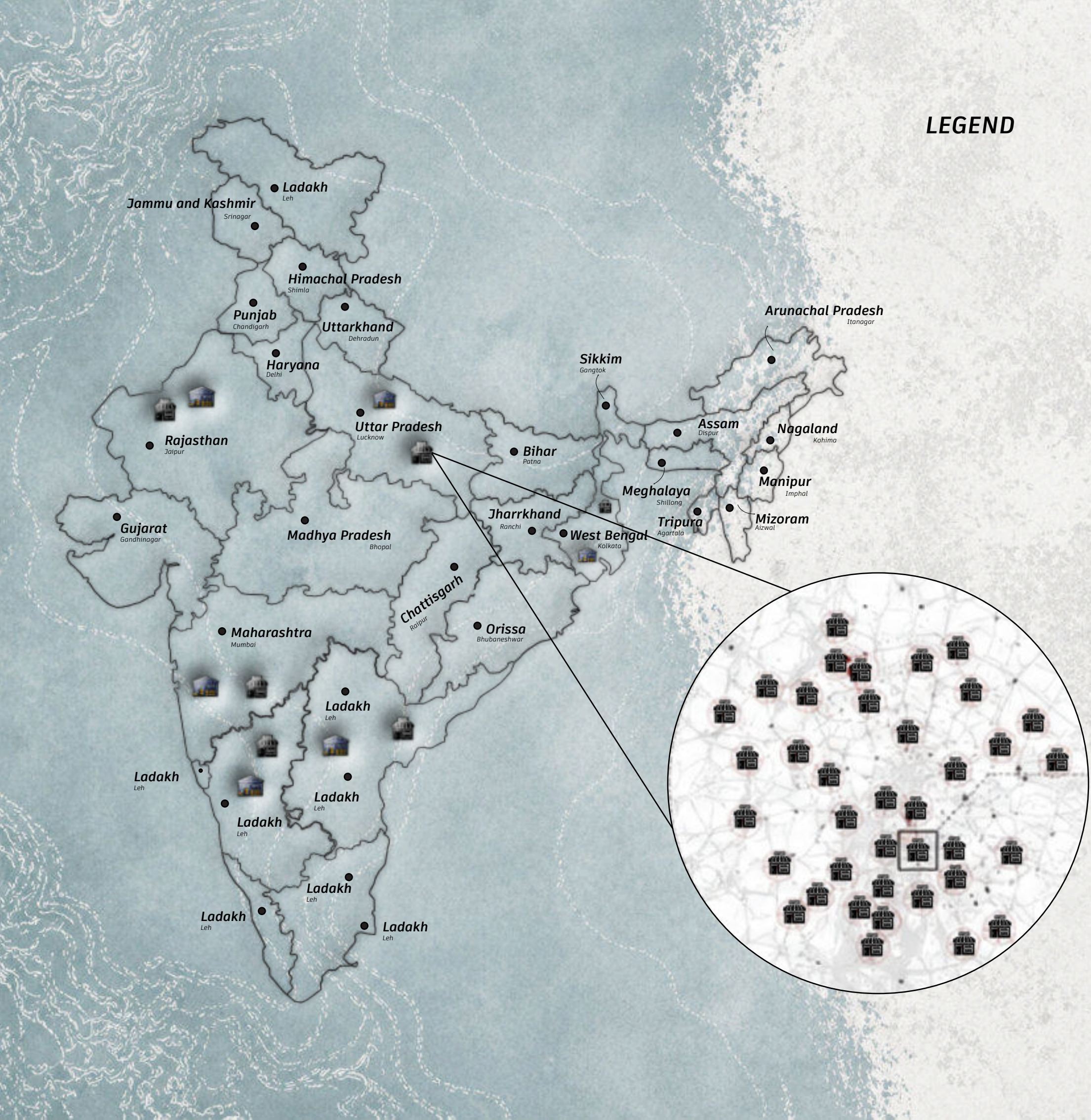
THE EXAMPLE SYSTEM

DECENTRALISED AND INSUFFICIENT

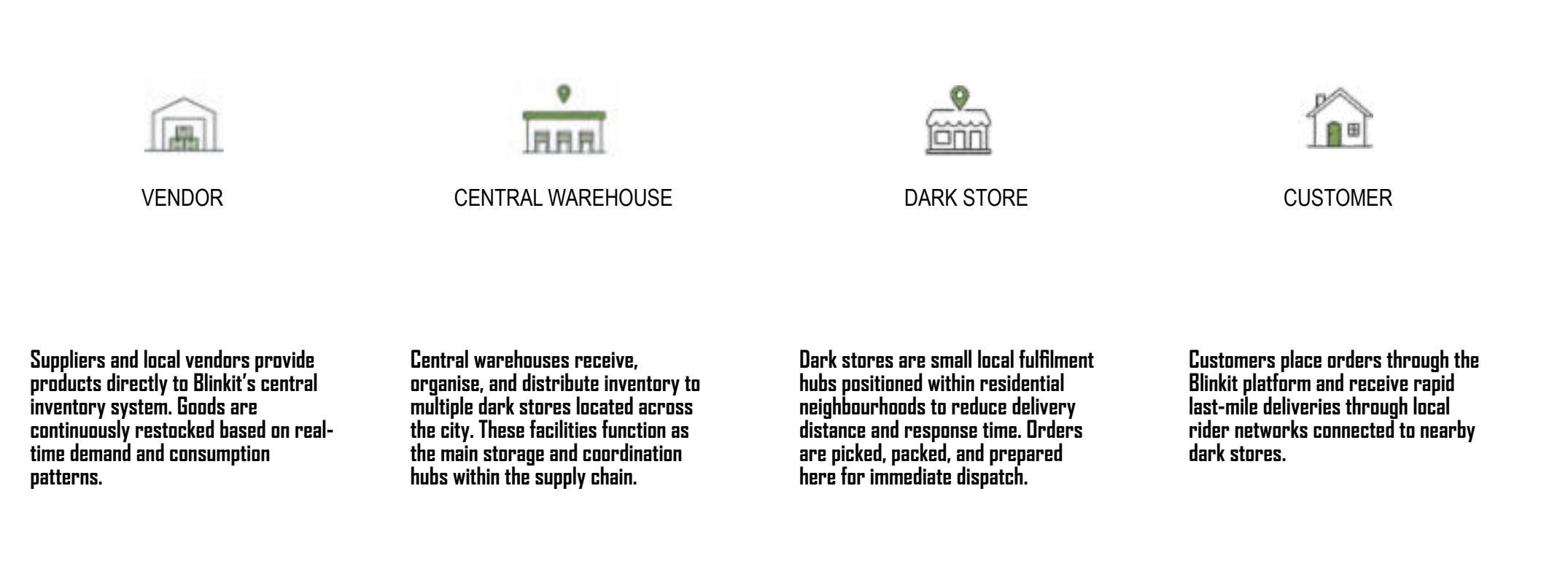
Blinkit is used in this project as a reference example because it demonstrates a fundamentally different approach to e-commerce logistics compared to the conventional warehouse-to-city delivery model. Instead of relying on a few large distribution centres located on the urban periphery, Blinkit operates through a highly decentralised quick-commerce system designed to minimise delivery time by distributing inventory across a dense network of small urban fulfilment centres known as “dark stores.” These micro-warehouses are embedded directly within residential neighbourhoods, allowing deliveries to be completed within minutes rather than hours or days.

Originally developed in India, Blinkit functions across major cities such as Delhi, Mumbai, Bangalore, Hyderabad, and Kolkata through hundreds of strategically placed dark stores connected to local rider networks and real-time inventory management systems. This hyper-local logistics model prioritises proximity over scale, enabling rapid delivery of groceries, household goods, and daily essentials while significantly reducing long-distance last-mile routes. Each dark store serves a limited delivery radius, creating an interconnected urban distribution network capable of responding quickly to local demand patterns and consumer behaviour.

The Blinkit system is particularly relevant to this project because it reveals the spatial and infrastructural implications of integrating logistics directly into the urban fabric. By decentralising storage and distribution, it offers an alternative framework for rethinking how future e-commerce systems could operate in dense cities such as Prague, where logistics currently remain dependent on peripheral warehouses and fragmented delivery networks. At the same time, the model also exposes new urban challenges, including increased rider traffic, operational intensity, labour dependency, and the continuous occupation of urban space for logistics functions. Through this example, the project investigates how decentralised micro-fulfilment systems can transform delivery speed, spatial organisation, and the relationship between logistics infrastructure and the contemporary city.



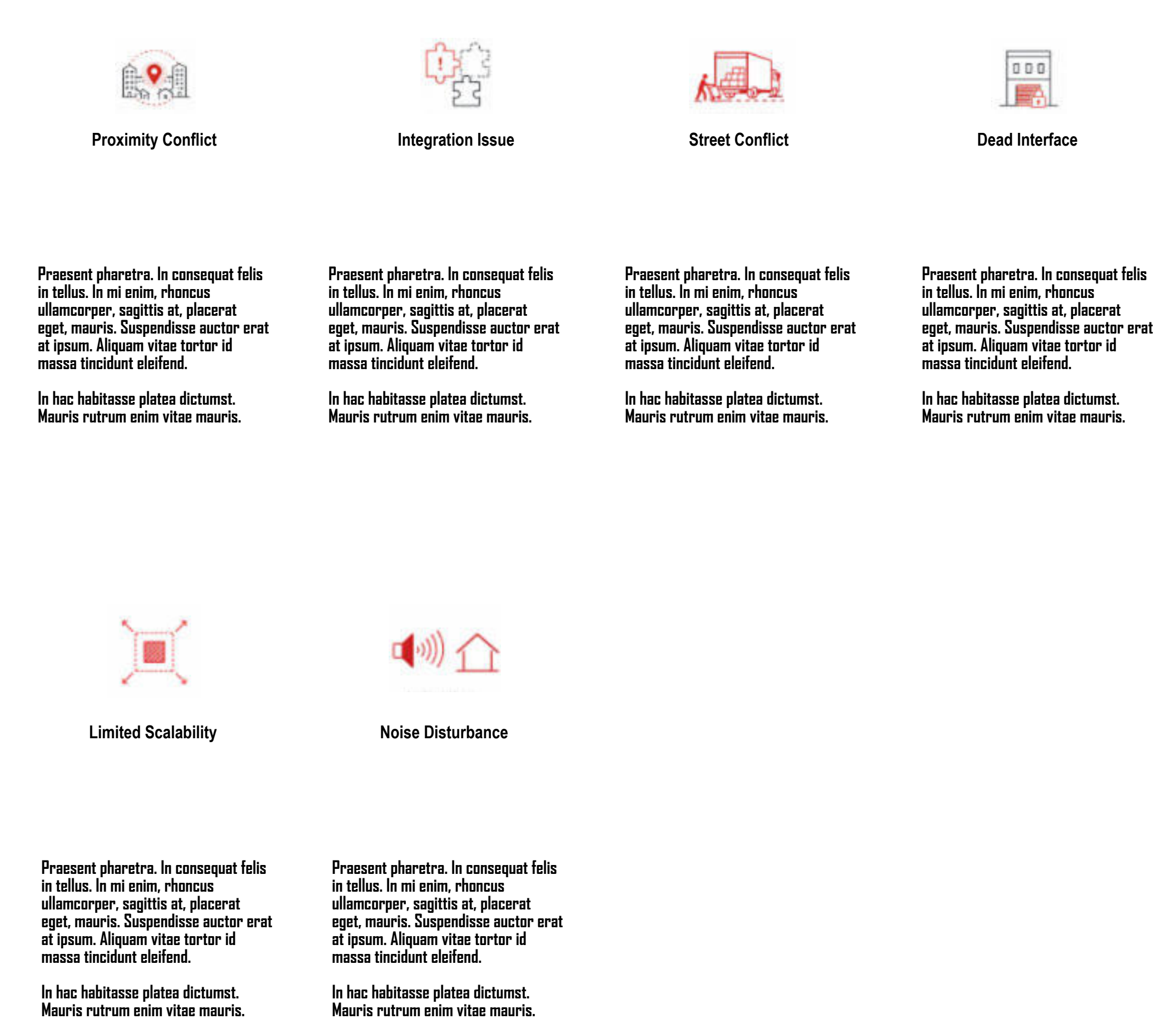
2.1 SUPPLY CHAIN OF THIS LOGISTIC



2.2 CONCEPT OF THE E-COMMERCE



2.3 PROBLEMS OF THIS LOGISTIC



THE CHANGED SYSTEM

DECENTRALISED AND INSUFFICIENT

The existing logistics system is increasingly shaped by the rapid growth of e-commerce and demand for immediate delivery services. In response, pickup lockers continue to spread across the city as isolated solutions to rising parcel volumes. However, this model remains insufficient in addressing long-term urban demand. The continuous multiplication of lockers fragments public space, increases infrastructural clutter, and fails to provide the speed and operational capacity required for high-frequency urban deliveries.

The proposed system introduces twelve strategically located microhubs operating as dark stores distributed throughout Prague. These hubs form a decentralised network designed to support high-speed local delivery through immediate response logistics. Rather than relying on endless expansion of pickup lockers, goods are consolidated, sorted, and dispatched directly from neighbourhood-scale fulfilment centres positioned closer to residential density and demand clusters.

The green delivery zones shown on the map represent a service radius of approximately 2–3 kilometres surrounding each microhub. This distance enables rapid delivery times, reduced travel distances, and faster response capacity within dense urban areas. By positioning dark stores within close proximity to consumers, the system supports same-hour and immediate delivery models while minimising dependency on long-distance transportation from peripheral warehouses.

Each microhub functions as an active local distribution node capable of handling storage, sorting, and dispatch operations independently. Together, the twelve hubs create overlapping delivery coverage across the metropolitan area, ensuring continuous accessibility while reducing logistical pressure on the city centre. The overlap between service zones strengthens redundancy and flexibility within the network, allowing deliveries to adapt dynamically according to demand fluctuations.

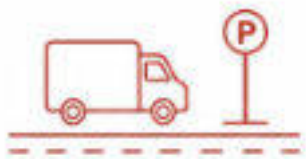


3.1 SUPPLY CHAIN TRANSFORMATION

WHAT WE REMMOVE



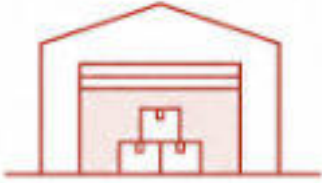
USER



APP INTERFACE



DARK STORE

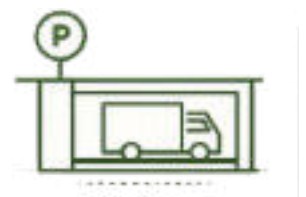


DELIVERY

WHAT WE ADD



USER



APP INTERFACE

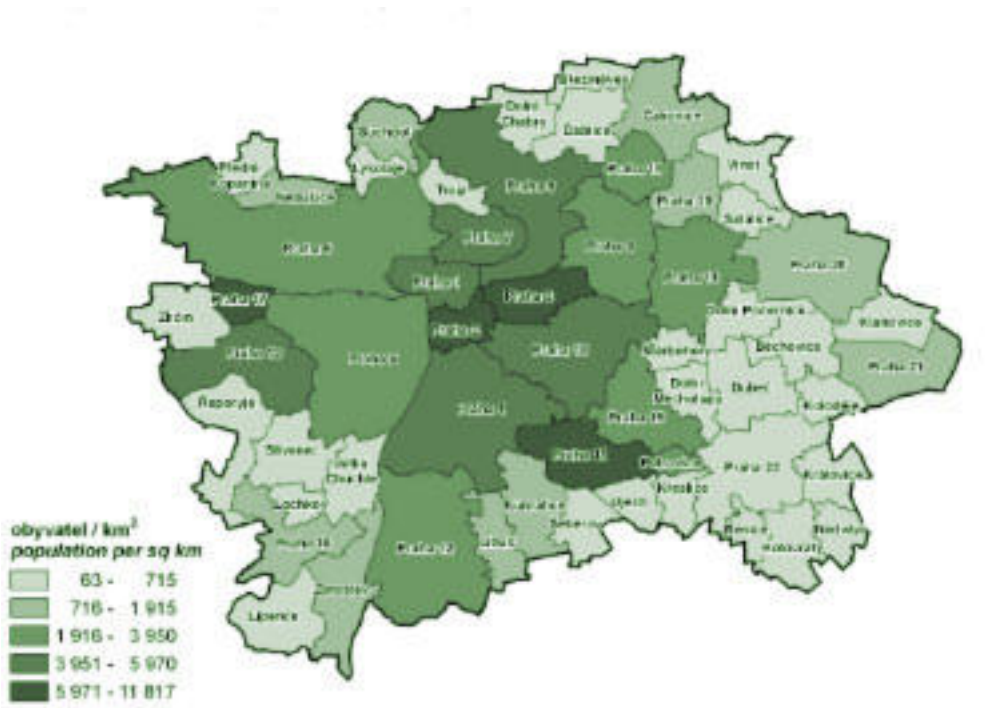


DARK STORE

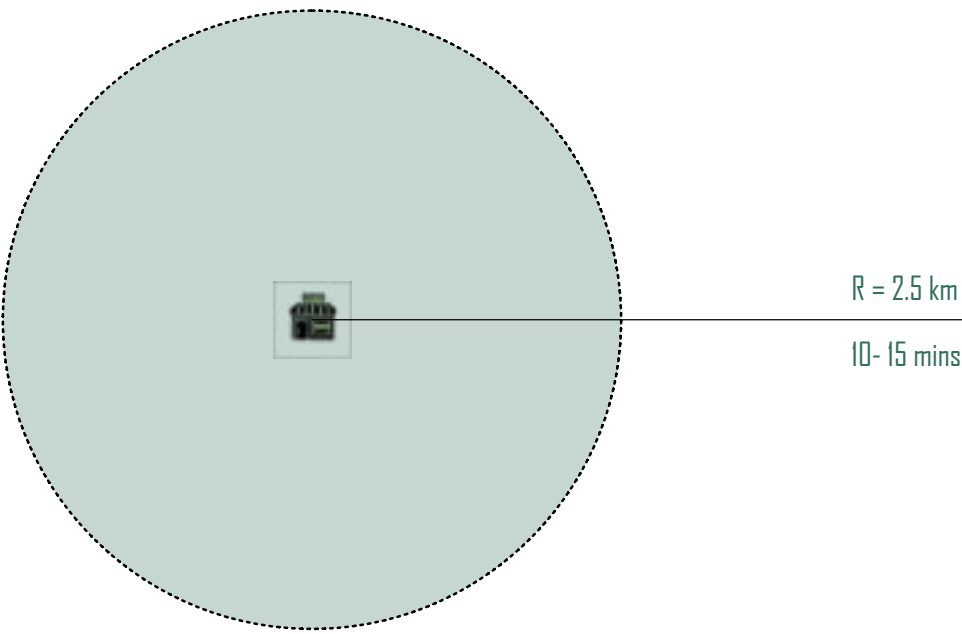


DELIVERY

3.2 DARK STORE PLACEMENT LOGIC



A shaded regional map visualizing population or demand density across different districts. Darker green areas indicate higher concentration zones, helping identify optimal locations for dark store placement based on accessibility and demand intensity.



A circular service-area diagram illustrating the operational reach of a dark store. The center represents the warehouse location, while the surrounding radius highlights a 2.5 km delivery coverage area with an estimated delivery time of approximately 5 minutes.

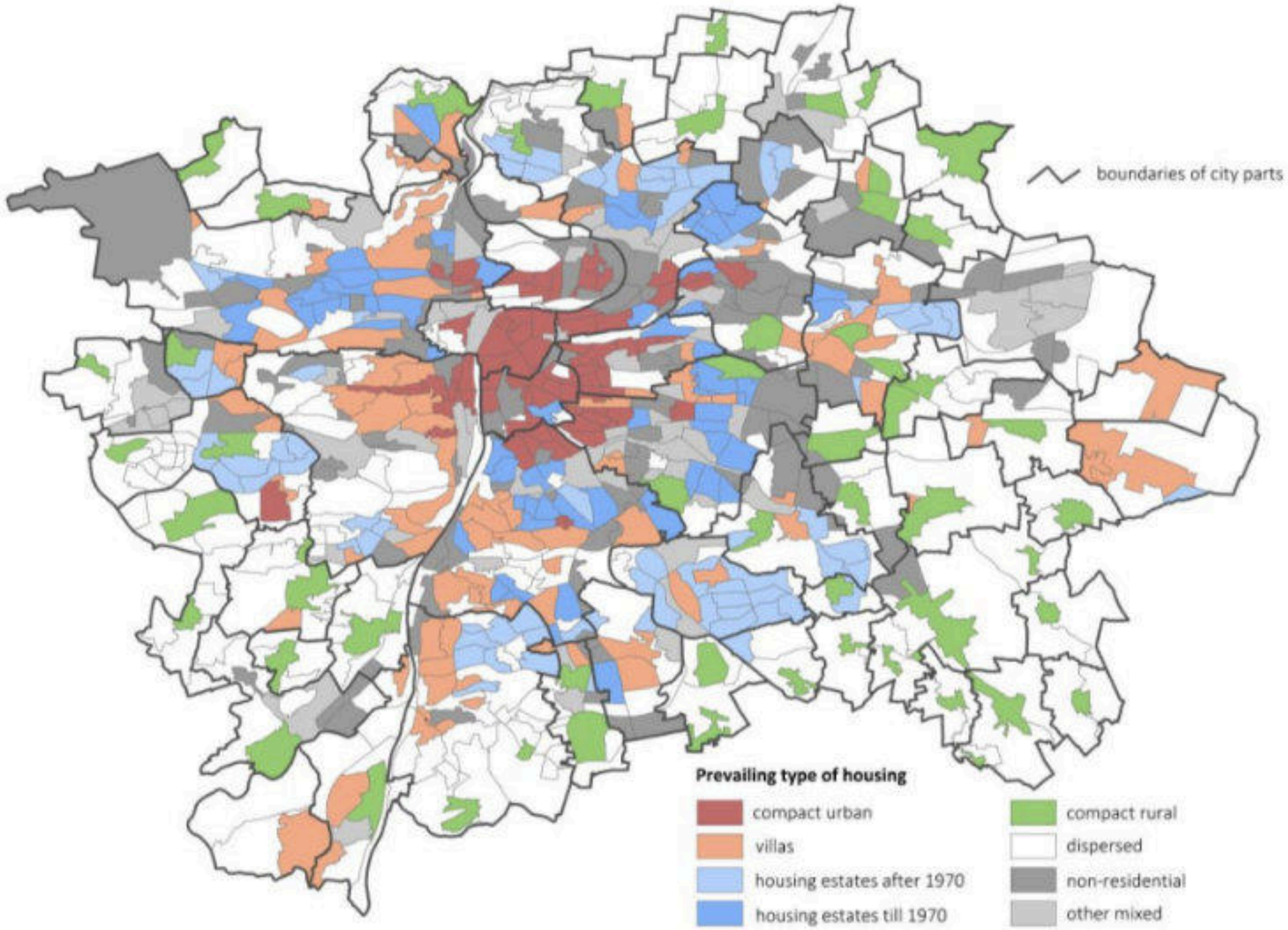
PRAGUE URBAN STUDY

The urban study examines the spatial structure and housing distribution of Prague in order to identify suitable conditions for decentralized logistics infrastructure. The city is characterized by a highly diverse urban fabric composed of compact historical districts, post-war housing estates, suburban residential areas, and dispersed peripheral developments. This diversity creates varying logistical demands and different levels of accessibility across the metropolitan territory.

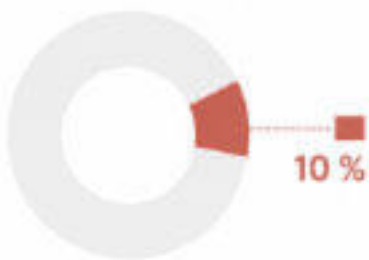
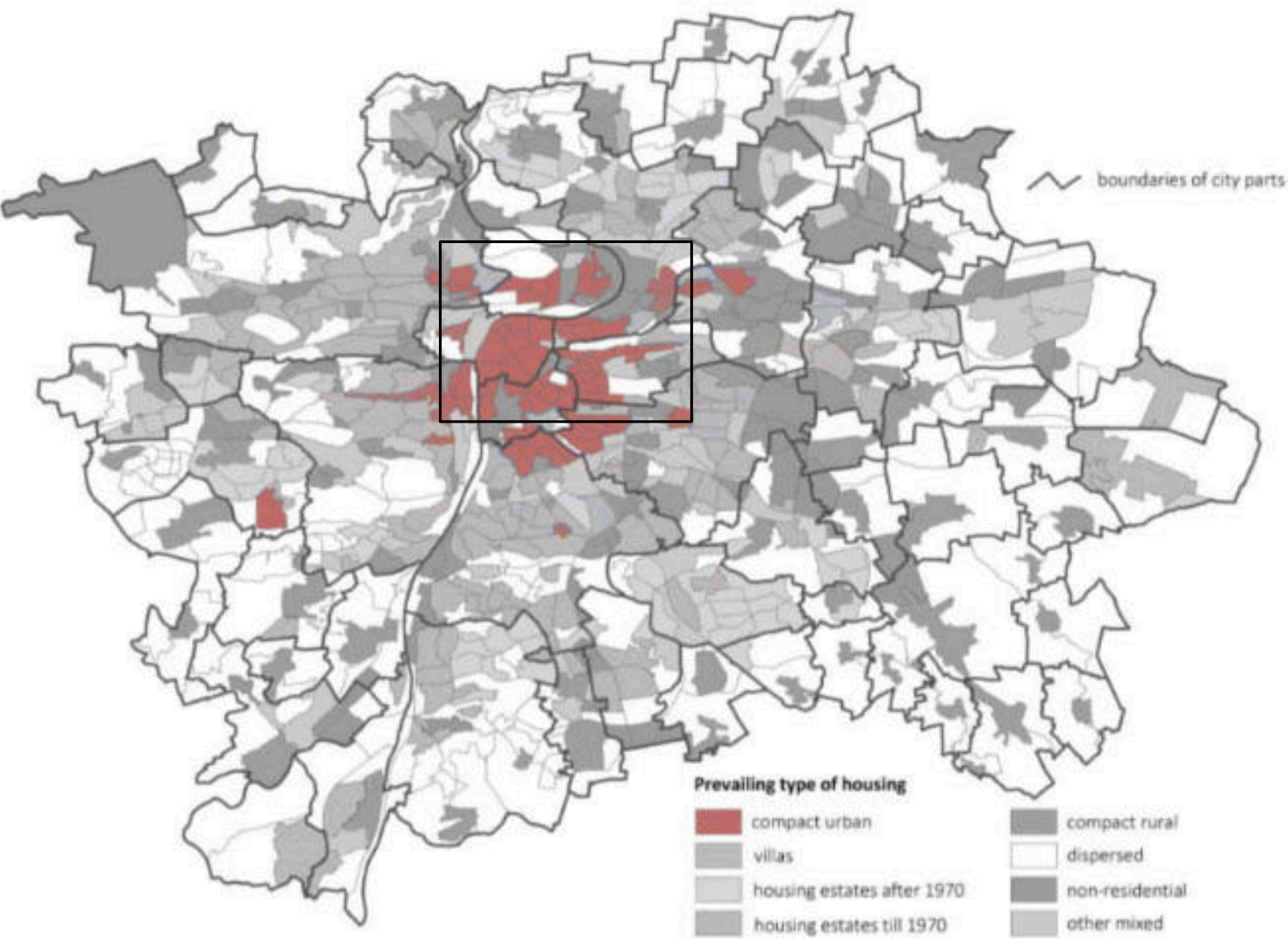
The analysis highlights the concentration of dense mixed-use neighborhoods in the central areas of Prague, where high population density and increasing delivery activity place pressure on existing transport systems and public space. At the same time, large residential housing estates from the twentieth century form significant urban clusters with strong local demand but limited proximity to contemporary distribution infrastructure. Peripheral villa districts and suburban developments present additional challenges related to longer delivery routes and dependence on road transport.

Understanding these urban patterns is essential for defining a network of localized storage and distribution points. The study suggests that decentralized logistics can operate most effectively when strategically inserted within residential zones that combine accessibility, population density, and available residual space. Rather than concentrating logistics functions in isolated industrial zones, the proposal advocates for a distributed system integrated throughout the city structure.

The mapping of housing typologies also reveals opportunities for adapting logistics infrastructure to different urban contexts. Compact inner-city districts may require highly integrated and architecturally sensitive solutions, while larger residential estates can accommodate shared neighborhood facilities with broader public interfaces. Through this analysis, Prague is understood not only as a transportation network, but as a layered urban environment where logistics, housing, mobility, and public life intersect.



COMPACT URBAN



GENERAL SITE ANALYSIS



Spatial Structure

Dense perimeter-block buildings form enclosed courtyards and continuous street edges, creating a compact and walkable urban fabric.

Main Streets & Connectivity

Primary streets carry tram lines, retail activity, and high pedestrian movement, while secondary streets provide quieter residential access.

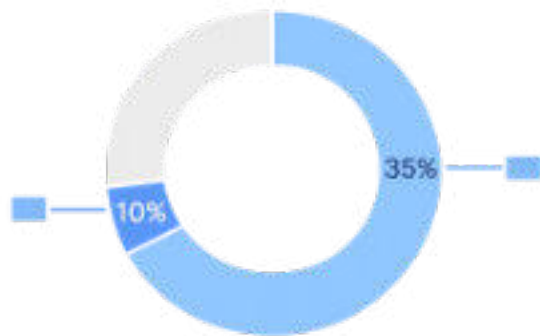
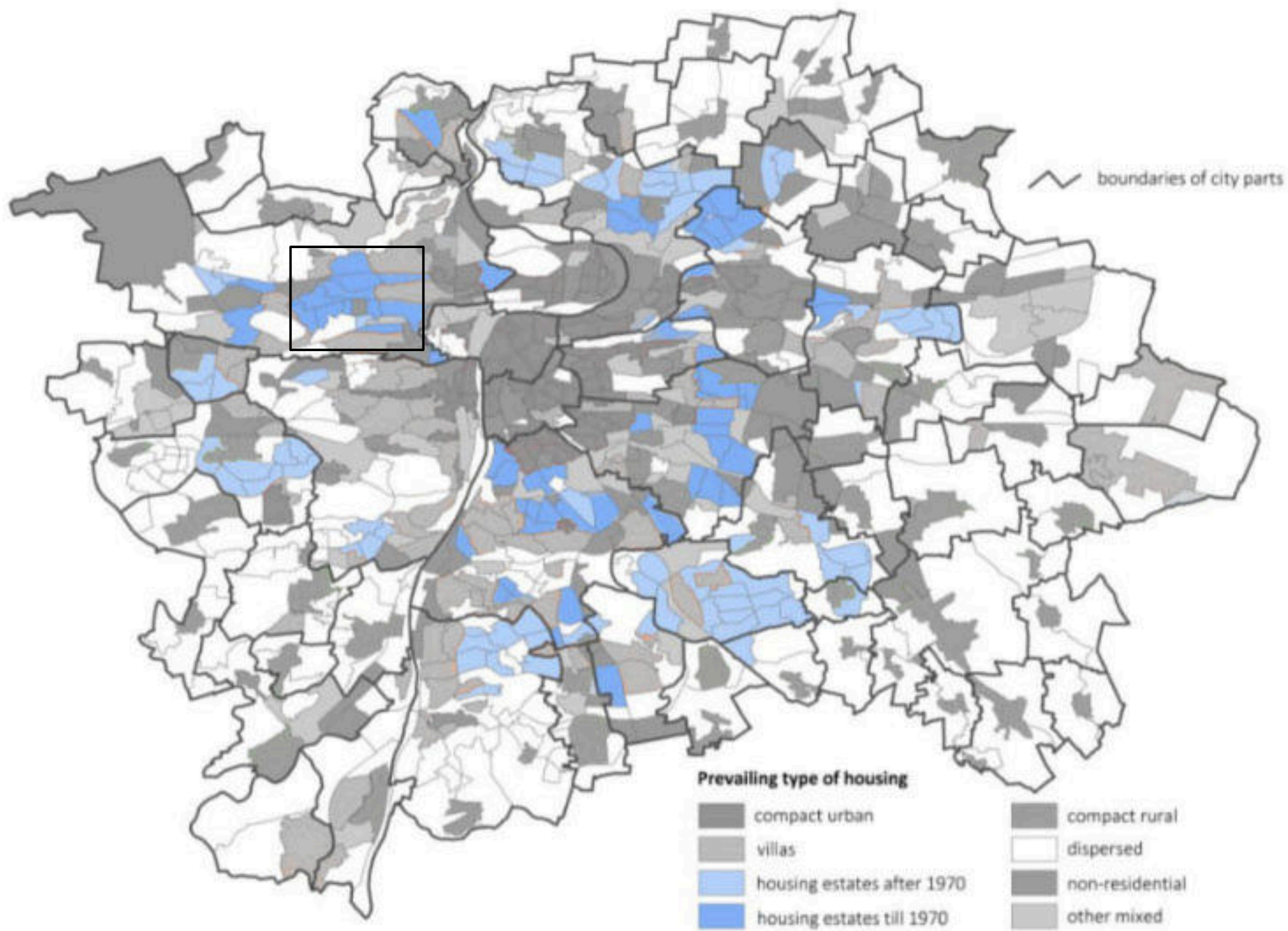
Green Space

Greenery is mainly located in internal courtyards and small public squares, integrated within the block structure rather than separating buildings.

Land Use

The area is predominantly residential with mixed-use ground floors containing shops, cafés, services, and some offices along main streets.

HOUSING ESTATES



GENERAL SITE ANALYSIS



Spatial Structure

Freestanding slab buildings are arranged within large open spaces, creating a loose and repetitive urban layout with separated building clusters.

Main Streets & Connectivity

Wide vehicular roads connect the estates to the city, while internal pedestrian pathways provide access between housing blocks, schools, and local centers.

Green Space

Large shared green areas and parks are located between buildings, forming an open landscape structure rather than enclosed courtyards.

Land Use

The estates are predominantly residential, with shops, schools, and services concentrated in central nodes rather than integrated continuously along streets.

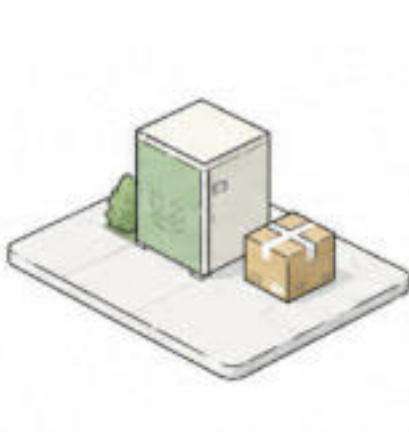
DESIGN CONCEPT

The project proposes a decentralized urban logistics system that integrates small-scale storage infrastructure directly into the public fabric of the city. Instead of relying solely on large peripheral warehouses, the concept introduces compact neighborhood storage units positioned closer to residential areas, reducing delivery distances and improving last-mile efficiency. These micro-facilities are designed not as isolated technical objects, but as hybrid urban elements that contribute positively to public space.

The design approach focuses on combining logistical function with civic value. Storage volumes are embedded into existing urban conditions and paired with publicly accessible interfaces such as seating, greenery, bicycle parking, waste collection, and small communal spaces. By softening the visual and spatial impact of logistics infrastructure, the intervention aims to make distribution systems more compatible with everyday urban life while improving accessibility for residents.

A modular architectural language allows the system to adapt to different scales and contexts throughout the city. Units can be implemented in dense urban blocks, residential neighborhoods, or transitional areas while maintaining a consistent identity. Their compact footprint enables flexible placement within underused parcels, parking lots, street edges, or residual urban spaces. This distributed strategy supports more resilient and sustainable logistics networks by shortening transportation routes, minimizing traffic pressure, and encouraging alternative delivery methods such as cycling or electric mobility.

The proposal ultimately reframes logistics infrastructure as part of the public realm rather than hidden back-of-house service space. By integrating storage with social and environmental functions, the concept creates a multifunctional urban typology that responds simultaneously to growing delivery demands, spatial efficiency, and the quality of everyday city life.



Storage for goods



Closer to people within the city



Add public user interface
(seating, planter, bin, etc.)



Integrated solution combining
storage and public use

DESIGN PROCESS



Base Geometry



Protective Outer Surface



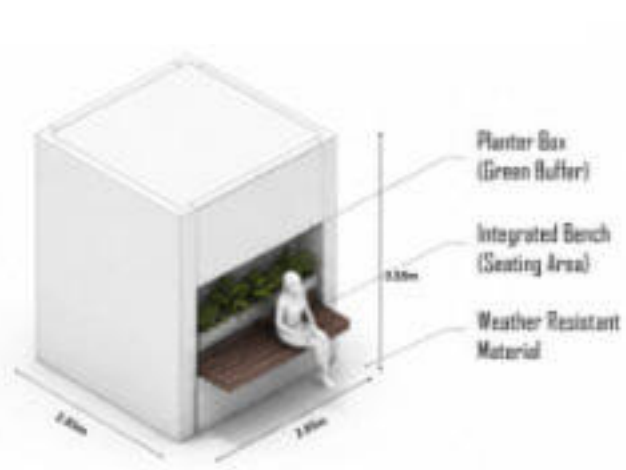
Pickup Counter



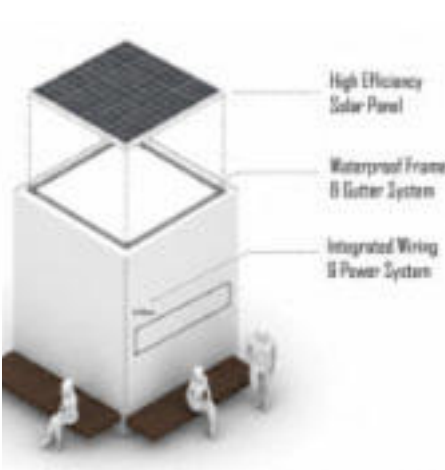
Vertical Scalability



Horizontal Scalability



Solar Panels



Public User Interface

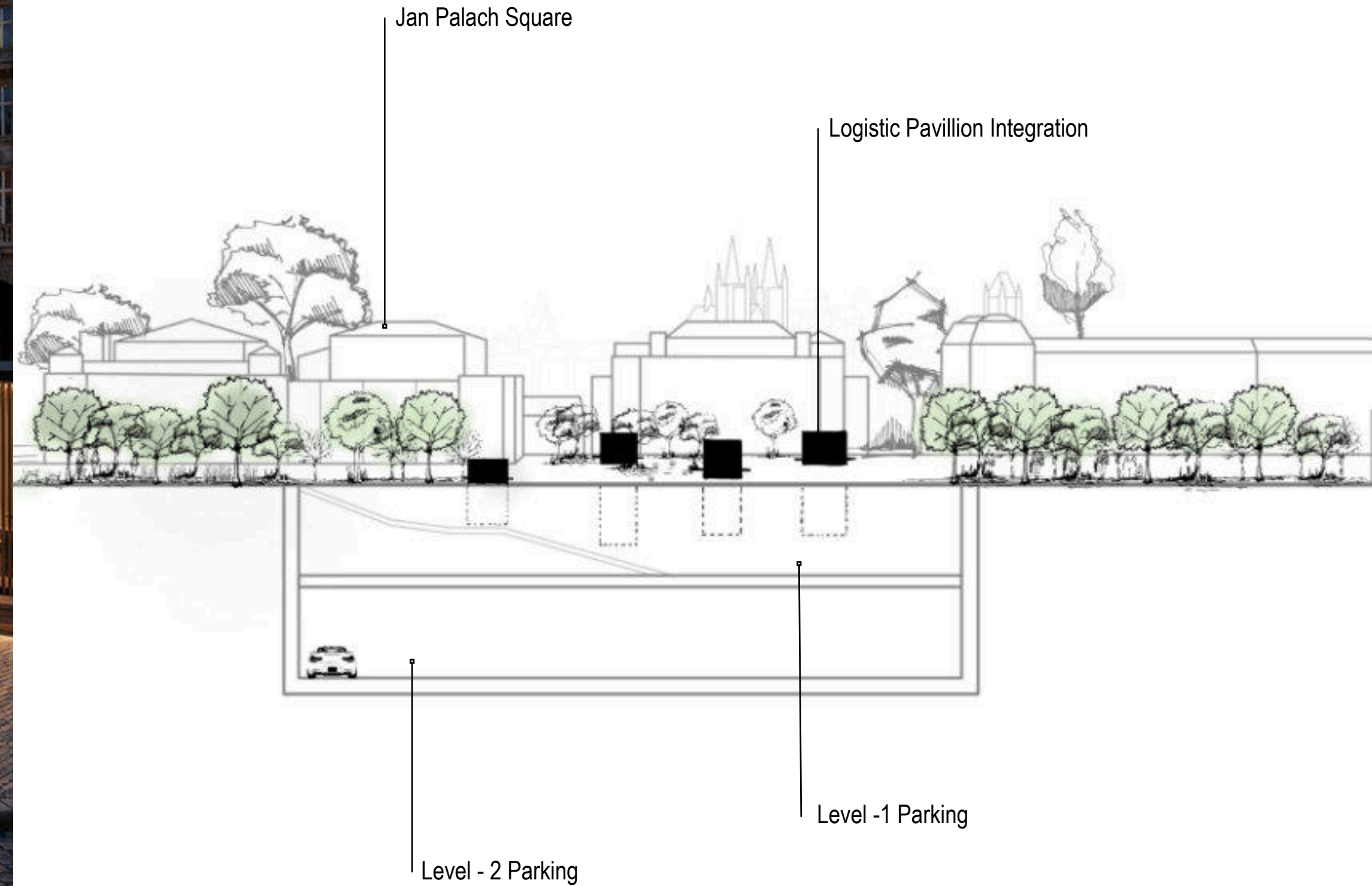


Urban Logistic PlugIn

DESIGN INTERGATION IN COMAPCT URBAN



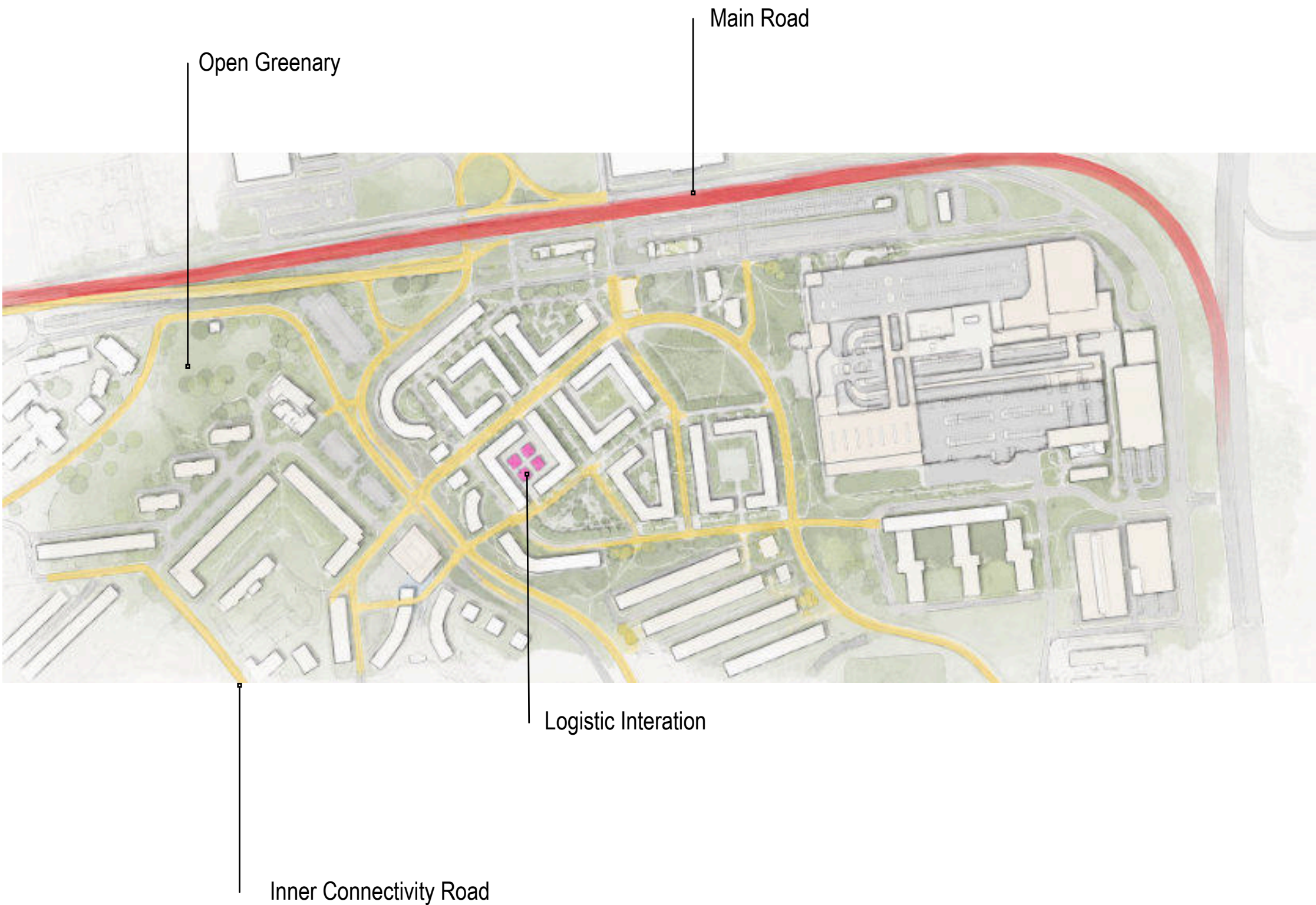
A compact urban logistics pavilion seamlessly integrated into the historic city fabric, combining underground parking and concealed delivery infrastructure while preserving public space, walkability, and the architectural character of the square.



DESIGN INTEGRATION IN PANELAK



A community-focused logistics hub integrated within a residential panel housing district, activating open public spaces through mixed-use functions, local markets, and improved inner connectivity while supporting efficient neighborhood-scale delivery systems.



URBAN KPI

Social Interaction Index	Users/day in courtyard
Public Safety Activation	% increase in active public space

BEFORE AND AFTER



GREENNARY KPI

Green Space Preservation	% courtyard maintained as green
Green Space Activation	Users/day using green area

BEFORE AND AFTER



ENERGY KPI

Renewable Energy Contribution	% hub energy from solar
Energy Shared with Residences	kWh/year exported

BEFORE AND AFTER



